

Naomi SCM (Supply Chain Management)- Work Distribution Document

Project Overview

Project Name: Naomi SCM - Factory PO Management System

Version: 2.0.0

Document Date: 2026-01-21

Status: Requirements Frozen, Demo Completed, Development Pending

Table of Contents

1. [Application Pages](#)
 2. [UI Work Items](#)
 3. [API Work Items](#)
 4. [Database Work Items](#)
 5. [Email Alert System](#)
 6. [Work Distribution Summary](#)
-

1. Application Pages

1.1 Login Page

- User authentication with username/password
- Role-based session management
- HTTP-only cookie authentication
- 5 user roles: NH_LEAD, NH_OPS, NH_USOPS, FACTORY_AGENT, WH_AGENT

1.2 PO Creation Page

- Create new Purchase Order
- Factory and Warehouse selection (dropdowns from master data)
- Cost Basis selection

- Requested Ship Date and ETA date pickers
- SKU item entry with autocomplete search
- Add/Remove SKU line items
- Quantity and pricing per SKU
- Total value calculation
- Save as Draft / Submit for Approval

1.3 PO Listing Page

- Role-based PO listing with different columns per role
- Filters: Search, Status, Warehouse, Factory, Date Range
- Pagination (25 items per page)
- Bulk Actions:
 - NH_LEAD: Bulk Approve (Open, Pending Approval status)
 - NH_OPS: Bulk Submit for Approval (Open status)
- Preview button to navigate to workflow

1.4 PO Workflow Page (Tabbed Interface)

Tab	Event ID	Description
Modification	EVENT2	FPO details, Customs Estimate, Item Details, Approval workflow
Production	EVENT3	Production Tracker, Loading Confirmation, Logistics Handoff
Logistics	EVENT4	Shipping details, Telex documents, Warehouse tracking
Inventory	EVENT5	Received quantities, Damage tracking, Inventory confirmation

1.5 Settings Page (NH_LEAD only)

- RBAC configuration matrix
- Event access management per role
- Protected roles and events handling

2. UI Work Items

Note: UI was demonstrated as a prototype. Actual development pending based on demo.

2.1 Common Components (Build First)

#	Work Item	Status	Est. Hours
UI-001	Common Components (sidebar, header, toast, modals, spinners, form validation)	Pending	12

2.2 Login Page

#	Work Item	Status	Est. Hours
UI-002	Login Page (form, validation, error display, session management, redirect)	Pending	12

2.3 PO Creation Page

#	Work Item	Status	Est. Hours
UI-019	PO Creation Page (factory/warehouse/cost basis dropdowns, date pickers, SKU autocomplete, add/remove items, totals, save draft/submit)	Pending	12

2.4 PO Listing Page

#	Work Item	Status	Est. Hours
UI-003	List Table Card (role-based columns, NH Team/Factory Agent/WH Agent tables, pagination)	Pending	8
UI-004	Filters Card (search, status, warehouse, factory, date range, clear filters)	Pending	8
UI-005	Bulk Actions Card (select all, checkboxes, bulk approve NH_LEAD, bulk submit NH_OPS)	Pending	6

2.5 PO Workflow - Modification Tab (EVENT2)

#	Work Item	Status	Est. Hours
UI-006	PO Details Card (factory, warehouse, cost basis, ship date, ETA, total value)	Pending	6
UI-007	Customs Estimate Card (ocean freight, drayage, chassis, custom duties with DDP hide)	Pending	6

#	Work Item	Status	Est. Hours
UI-008	Item Details Card (SKU autocomplete, add/remove rows, totals, save/submit, approve/reject, remarks)	Pending	8

2.6 PO Workflow - Production Tab (EVENT3)

#	Work Item	Status	Est. Hours
UI-009	Production Tracker Card (sequential field unlock, factory agent approval, hold PO toggle)	Pending	6
UI-010	Loading Confirmation Card (SKU table, loading qty editable, row/footer totals)	Pending	8
UI-011	Logistics Handoff Card (container number, actual ship date, update button)	Pending	6

2.7 PO Workflow - Logistics Tab (EVENT4)

#	Work Item	Status	Est. Hours
UI-012	Shipping Information Card (carrier, forwarder, vessel, terminal, HBL, MBL, freight rate)	Pending	6
UI-013	Telex Documents Card (upload, preview, download, delete)	Pending	8
UI-014	Warehouse Tracking Card (vessel status, LFD, per diem, rail dates, pickup, delivery)	Pending	6
UI-015	Empty Container Card (notification date, pickup date, return date)	Pending	6

2.8 PO Workflow - Inventory Tab (EVENT5)

#	Work Item	Status	Est. Hours
UI-016	SKU Items Card (item cards, loading qty, received/damaged qty, missing/usable calculation)	Pending	8
UI-017	Summary Card (totals, inventoried date, confirm receipt button)	Pending	8

2.9 Settings Page

#	Work Item	Status	Est. Hours
UI-018	RBAC Settings Card (matrix table, role checkboxes, protected indicators, save button)	Pending	12

3. API Work Items

3.1 Authentication APIs

#	Endpoint	Method	Status	Est. Hours
API-001	/auth/login	POST	Pending	
API-002	/auth/logout	POST	Pending	

3.2 Master Data APIs

#	Endpoint	Method	Status	Est. Hours
API-003	/master/factories	GET	Pending	6
API-004	/master/warehouses	GET	Pending	6
API-005	/master/cost-basis	GET	Pending	6
API-006	/master/forwarders	GET	Pending	6
API-007	/master/carriers	GET	Pending	6
API-008	/master/origin-ports	GET	Pending	6
API-009	/master/vessel-status	GET	Pending	6
API-010	/master/vessel-status	POST	Pending	0
API-011	/master/sku	GET	Pending	6
API-012	/master/rbac-rules	GET	Pending	0

3.3 PO Management APIs

#	Endpoint	Method	Status	Est. Hours
---	----------	--------	--------	------------

#	Endpoint	Method	Status	Est. Hours
API-013	/po/create	POST	Pending	12
API-014	/po/list	POST	Pending	12
API-015	/po/details	POST	Pending	0
API-016	/po/bulk-approve	POST	Pending	12
API-017	/po/bulk-submit	POST	Pending	12

3.4 PO Workflow APIs

#	Endpoint	Method	Status	Est. Hours
API-018-1	/po/event?#modification	GET	Pending	8
API-018-2	/po/event?#production	GET	Pending	8
API-018-3	/po/event?#logistics	GET	Pending	8
API-018-4	/po/event?#inventory	GET	Pending	8
API-019	/po/modification	PUT	Pending	8
API-020	/po/production	PUT	Pending	8
API-021	/po/logistics	PUT	Pending	8
API-022	/po/inventory	PUT	Pending	8
API-023	/po/history	GET	Pending	8

3.5 File Management APIs

#	Endpoint	Method	Status	Est. Hours
API-024	/po/telex/upload	POST	Pending	8
API-025	/po/telex/download	POST	Pending	8
API-026	/po/telex/delete	POST	Pending	8

3.6 Settings APIs

#	Endpoint	Method	Status	Est. Hours
API-027	/settings	GET	Pending	6

#	Endpoint	Method	Status	Est. Hours
API-028	/settings	PUT	Pending	6

4. Database Work Items

4.1 Core Tables

#	Table Name	Status	Est. Hours
DB-001	users	Pending	
DB-002	sessions	Pending	
DB-003	purchase_orders	Pending	
DB-004	po_items	Pending	
DB-005	production_tracker	Pending	
DB-006	logistics_tracker	Pending	
DB-007	inventory_tracker	Pending	
DB-008	purchase_order_transactions	Pending	
DB-009	po_history	Pending	
DB-010	telex_documents	Pending	

4.2 Master Data Tables

#	Table Name	Status	Est. Hours
DB-011	factories	Pending	
DB-012	warehouses	Pending	
DB-013	sku_master	Pending	
DB-014	forwarders	Pending	
DB-015	carriers	Pending	
DB-016	origin_ports	Pending	
DB-017	cost_basis	Pending	
DB-018	vessel_status	Pending	

#	Table Name	Status	Est. Hours
DB-019	events	Pending	

4.3 RBAC Tables

#	Table Name	Status	Est. Hours
DB-020	rbac_settings	Pending	
DB-021	rbac_field_permissions	Pending	

5. Email Alert System

5.1 Email Alert Requirements

Alert ID	Condition	Trigger Type	Est. Hours
EMAIL-001	Expected Production Start Date not updated (72 hours)	Scheduled	
EMAIL-002	Ready Date not updated (72 hours)	Scheduled	
EMAIL-003	Container arriving to port (15 days before)	Scheduled	
EMAIL-004	Container arrived at Port	Event-triggered	
EMAIL-005	Pickup date not entered (LFD - 1 day)	Scheduled	
EMAIL-006	Delivered but not inventoried (48 hours)	Scheduled	
EMAIL-007	Empty date not entered (Per Diem - 1 day)	Scheduled	
EMAIL-008	Any field re-updated/modified	Event-triggered	
EMAIL-009	NH_LEAD selects Modification Required	Event-triggered	
EMAIL-010	NH_LEAD selects Reject PO	Event-triggered	

5.2 Email Alert Details

Note: Recipients for all email alerts are yet to be confirmed. Sample warehouse-specific recipients:

- **FCI Warehouse:** SWalston@ssco.pro, JYamaguchi@ssco.pro, dsandifer@ssco.pro, etc.
- **ARMS Warehouse:** scho@arms-logistics.com, jwang@arms-logistics.com, etc.
- **CC (All Alerts):** nh_po@ojcommerce.com, naomihome@ojcommerce.com

EMAIL-001: Expected Production Start Date Not Updated

Trigger: Daily scheduler checks POs in "Approved" or "Agent Approved" status

Condition: expectedProductionStart is NULL or empty for > 72 hours after approval

Subject: Production Start Date / Ready Date not updated within 72 hours

Body Template:

Hi Team,

This is a reminder that the Production Start Date has not been updated within 72 hours for the following PO's. Please update ASAP.

PO Number Warehouse Factory PO Receive Date
----- ----- ----- -----
{poNumber} {warehouse} {factory} {poReceiveDate}

Best,

Naomi Home Team

EMAIL-002: Ready Date Not Updated

Trigger: Daily scheduler checks POs in "In Production" status

Condition: readyDate is NULL or empty for > 72 hours after production started

Subject: Production Start Date / Ready Date not updated within 72 hours

Body Template:

Hi Team,

This is a reminder that the Ready Date has not been updated within 72 hours for the following PO's. Please update ASAP.

PO Number Warehouse Factory Production Start Date
----- ----- ----- -----
{poNumber} {warehouse} {factory} {productionStartDate}

---|-----|-----|-----|
| {car} | {customs} | {forwarder} | {carrier} | {doReceived} | {vessel} |
{terminal} | {bolNumber} | {containerNumber} | {arrivedDate} |

Best,
Naomi Home Team

EMAIL-005: Pickup Date Not Entered (LFD - 1 Day)

Trigger: Daily scheduler
Condition: LFD is set AND (LFD - current_date) == 1 day AND pickedDate is NULL
Subject: [URGENT] Pickup date not entered, LFD - 1

Body Template:

Hi Team,

URGENT: The pickup date has not been entered and the Last Free Day (LFD) is tomorrow for the following container(s). Please take immediate action to avoid demurrage/detention charges.

PO Number	Container#	LFD	Warehouse	Terminal
{poNumber}	{containerNumber}	{lfd}	{warehouse}	{terminal}

Best,
Naomi Home Team

EMAIL-006: Delivered But Not Inventoried (48 Hours)

Trigger: Scheduled (every 6 hours)
Condition: Status is "Arrived At WH" AND (current_time - warehouseDeliveryDate) > 48 hours
Subject: [ACTION REQUIRED] Container delivered 48 hours ago but not inventoried

Body Template:

Hi Team,

The following container(s) were delivered more than 48 hours ago but have not been inventoried yet. Please complete the inventory process ASAP.

PO Number	Container#	Warehouse	Delivery Date	Hours Since Delivery
-----	-----	-----	-----	-----
{poNumber}	{containerNumber}	{warehouse}	{deliveryDate}	{hoursSinceDelivery}

Best,
Naomi Home Team

EMAIL-007: Empty Date Not Entered (Per Diem - 1 Day)

Trigger: Daily scheduler
Condition: Per Diem end date is approaching (1 day) AND emptyContainerReturnDate is NULL
Subject: [URGENT] Empty date not entered, Per Diem - 1

Body Template:

Hi Team,

URGENT: The empty container return date has not been entered and the Per Diem deadline is tomorrow for the following container(s). Please take immediate action to avoid additional per diem charges.

PO Number	Container#	Per Diem End	Warehouse
-----	-----	-----	-----
{poNumber}	{containerNumber}	{perDiemEnd}	{warehouse}

Best,
Naomi Home Team

EMAIL-008: Field Re-Updated/Modified

Trigger: Event-triggered on any PUT request to /po/* endpoints
Condition: Any field value changed from previous value (tracked in po_history)
Subject: Field/Data Updated - PO: {poNumber}

Body Template:

Hi Team,

The following field has been updated for PO {poNumber}:

Field	Old Value	New Value	Updated By	Updated At
{fieldName}	{oldValue}	{newValue}	{updatedBy} ({role})	{timestamp}

Best,
Naomi Home Team

EMAIL-009: Modification Required

Trigger: Event-triggered when NH_LEAD selects "Modification Required" action
Subject: Modification Required - PO: {poNumber}

Body Template:

Hi Team,

Modifications have been requested for PO {poNumber}.

Requested By: {nhLeadName} (NH_LEAD)
Date: {requestDate}

Remarks/Reason:
{modificationRemarks}

Please review and make the necessary changes.

Best,
Naomi Home Team

EMAIL-010: Reject PO

Trigger: Event-triggered when NH_LEAD selects "Reject" action
Subject: PO Rejected - PO: {poNumber}

Body Template:

Hi Team,

PO {poNumber} has been rejected.

Rejected By: {nhLeadName} (NH_LEAD)
Date: {rejectDate}

Reason:
{rejectionRemarks}

Best,
Naomi Home Team

5.3 Email System Work Items

UI Work Items (HTML Email Templates with Placeholders)

#	Work Item	Alert	Placeholders	Status	Est. Hours
EMAIL-UI-001	production_date_missing.html	EMAIL-001	{poNumber}, {warehouse}, {factory}, {poReceiveDate}	Pending	6
EMAIL-UI-002	ready_date_missing.html	EMAIL-002	{poNumber}, {warehouse}, {factory}, {productionStartDate}	Pending	6
EMAIL-UI-003	container_arriving.html	EMAIL-003	{car}, {customs}, {forwarder}, {carrier}, {doReceived}, {vessel}, {terminal}, {bolNumber}, {containerNumber}, {eta}	Pending	6

#	Work Item	Alert	Placeholders	Status	Est. Hours
EMAIL-UI-004	container_arrived.html	EMAIL-004	{car}, {customs}, {forwarder}, {carrier}, {doReceived}, {vessel}, {terminal}, {bolNumber}, {containerNumber}, {arrivedDate}	Pending	6
EMAIL-UI-005	pickup_date_required.html	EMAIL-005	{poNumber}, {containerNumber}, {lfd}, {warehouse}, {terminal}	Pending	6
EMAIL-UI-006	inventory_pending.html	EMAIL-006	{poNumber}, {containerNumber}, {warehouse}, {deliveryDate}, {hoursSinceDelivery}	Pending	6
EMAIL-UI-007	empty_return_required.html	EMAIL-007	{poNumber}, {containerNumber}, {perDiemEnd}, {warehouse}	Pending	6
EMAIL-UI-008	field_modified.html	EMAIL-008	{poNumber}, {fieldName}, {oldValue}, {newValue}, {updatedBy}, {role}, {timestamp}	Pending	6
EMAIL-UI-009	modification_requested.html	EMAIL-009	{poNumber}, {nhLeadName}, {requestDate}, {modificationRemarks}	Pending	6
EMAIL-UI-010	po_rejected.html	EMAIL-010	{poNumber}, {nhLeadName}, {rejectDate}, {rejectionRemarks}	Pending	6

Note: UI team creates HTML email templates with placeholders. API team replaces placeholders with actual data and sends emails.

API Work Items

#	Work Item	Description	Status	Est. Hours
EMAIL-API-001	Email service integration	SMTP/SendGrid/SES configuration and connection	Pending	
EMAIL-API-002	Template placeholder replacement	Load HTML template, replace {placeholders} with actual data	Pending	
EMAIL-API-003	Scheduled job runner	Cron/scheduler for daily/hourly email checks	Pending	

#	Work Item	Description	Status	Est. Hours
EMAIL-API-004	Event-triggered email dispatcher	Send emails on status changes, field updates	Pending	
EMAIL-API-005	Email queue management	Queue emails, retry failed sends	Pending	
EMAIL-API-006	Email delivery tracking	Log sent emails, track delivery status	Pending	

Database Work Items

#	Work Item	Status	Est. Hours
EMAIL-DB-001	email_templates table	Pending	
EMAIL-DB-002	email_logs table	Pending	
EMAIL-DB-003	scheduled_alerts table	Pending	
EMAIL-DB-004	notification_config table	Pending	

6. Work Distribution Summary

6.1 Consolidated Estimates

Team	Items	Est. Hours	Notes
UI (Pages & Components)	19	152	Excludes Email Templates
UI (Email Templates)	10	60	HTML templates with placeholders
UI Total	29	212	
API (Core APIs)	28	204	Excludes duplicates (0 hrs)
API (Email System)	6	TBD	To be estimated
API Total	34	204+	
Database	25	TBD	To be estimated
Grand Total	88	416+	

Note: Items marked with 0 hours are duplicates (API-010, API-012, API-015) and excluded from totals.

6.2 By Category

Category	Total Items	Pending
UI	29	29
API	34	34
Database	25	25
Total	88	88

6.3 By Team

Frontend Team (UI)

- **Status:** Pending (Demo completed, actual development pending)
- **Items:** 29 work items (19 UI + 10 HTML Email Templates)
- **Estimated Hours:** 212 hours (152 UI + 60 Email Templates)
- **Reference:** Demo prototype available for UI specifications
- **Key Deliverables:**
 - Login Page (1 item)
 - PO Creation Page (1 item)
 - PO Listing (3 cards)
 - PO Workflow - Modification Tab (3 cards)
 - PO Workflow - Production Tab (3 cards)
 - PO Workflow - Logistics Tab (4 cards)
 - PO Workflow - Inventory Tab (2 cards)
 - Settings Page (1 card)
 - Common Components (1 item)
 - HTML Email Templates with placeholders (10 templates)

Backend Team (API)

- **Status:** Pending
- **Items:** 34 work items (28 APIs + 6 Email System)
- **Estimated Hours:** 204+ hours (Core APIs, Email System TBD)
- **Key Deliverables:**
 - Authentication APIs (2)
 - Master Data APIs (10)
 - PO Management APIs (5)
 - PO Workflow APIs (6)
 - File Management APIs (3)

- Settings APIs (2)
- Email service integration (SMTP/SendGrid/SES)
- Template placeholder replacement (load HTML, replace {placeholders} with data)
- Scheduled job runner (cron/scheduler)
- Event-triggered email dispatcher
- Email queue management
- Email delivery tracking

Database Team

- **Status:** Pending
- **Items:** 25 work items (21 tables + 4 Email tables)
- **Key Deliverables:**
 - Core tables (10)
 - Master data tables (9)
 - RBAC tables (2)
 - Email/Notification tables (4)

6.4 Priority Matrix

Priority	Category	Work Items
P0 - Critical	Database	Core tables, Users, Purchase Orders
P0 - Critical	API	Authentication, PO List, PO Event Data
P1 - High	UI	Login Page, PO Creation, PO Listing, PO Workflow tabs
P1 - High	API	PO Workflow APIs, Bulk Actions
P1 - High	Email	Event-triggered alerts (EMAIL-004, 008, 009, 010)
P2 - Medium	API	Master Data APIs, Settings
P2 - Medium	Email	Scheduled alerts (EMAIL-001 to 007)
P3 - Low	API	History

7. Technical Reference

7.1 API Specification

See: [API_SPECIFICATION.md](#)

7.2 Database Schema

See: [APL_SPECIFICATION.md](#) - Appendix E

7.3 RBAC Configuration

See: [README.md](#) - Role Access Matrix

7.4 Demo Prototype Reference

- **Location:** Current project folder (Naomi SCM)
 - **Purpose:** UI/UX reference for development
 - **Note:** Prototype uses mock data - actual implementation requires API integration
-

Document Owner: Parthiban S

Last Updated: 2026-01-21

Status: Requirements Frozen, Development Pending